

COMP8026

Modelling Real-Time Behaviour with Statecharts

Assignment 1

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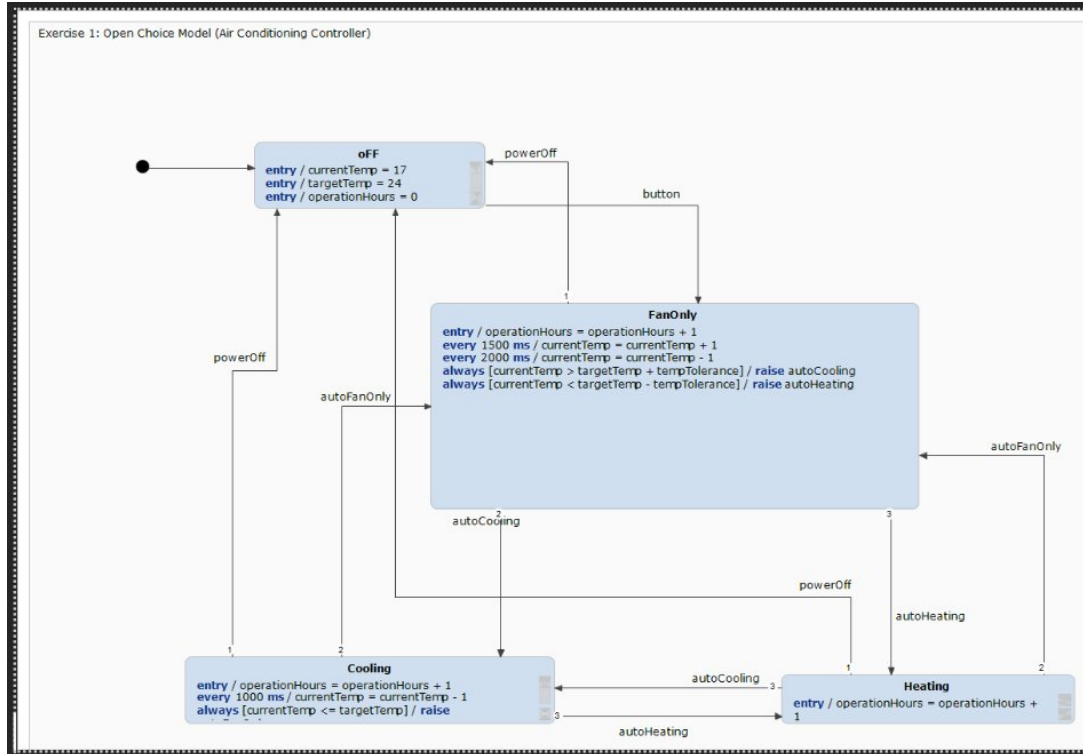
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COMP8026 Assignment 1

States and Stimuli Tables — All Exercises

Exercise 1: Open Choice Model — Air Conditioning Controller

Statechart Diagram:



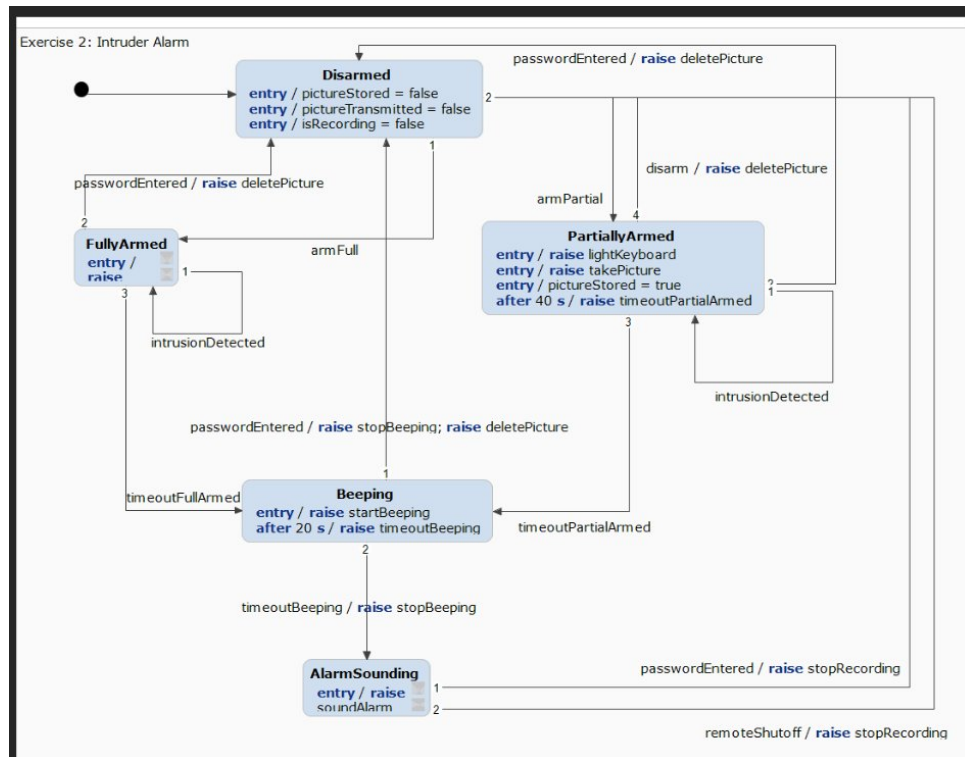
States and Stimuli:

State	Description
OFF	The initial state of the system. On entry, currentTemp is set to 17°C, targetTemp is set to 24°C, and operationHours is reset to 0. Transitions to FanOnly on a 'button' event, and can be reached from any active state via 'powerOff'.
FanOnly	The fan operates without heating or cooling. On entry, operationHours increments by 1. Every 1500 ms currentTemp increases by 1; every 2000 ms currentTemp decreases by 1. Raises 'autoCooling' if currentTemp > targetTemp + tempTolerance; raises 'autoHeating' if currentTemp < targetTemp - tempTolerance.
Cooling	Actively cools the environment. On entry, operationHours increments by 1. Every 1000 ms, currentTemp decreases by 1. Raises 'autoFanOnly' when currentTemp <= targetTemp. Can transition to Heating via 'autoHeating', to FanOnly via 'autoFanOnly', or to OFF via 'powerOff'.
Heating	Actively heats the environment. On entry, operationHours increments by 1. Raises 'autoFanOnly' when currentTemp reaches targetTemp. Can transition to Cooling via 'autoCooling', to FanOnly via 'autoFanOnly', or to OFF via 'powerOff'.
button	Stimulus. User presses the power button. Transitions the system from OFF to FanOnly, starting normal operation.
powerOff	Stimulus. Power-off command received. Transitions from any active state (FanOnly, Cooling, or Heating) back to OFF, resetting all parameters on entry.

State	Description
autoCooling	Stimulus. Raised automatically when $\text{currentTemp} > \text{targetTemp} + \text{tempTolerance}$. Transitions from FanOnly to Cooling, or from Heating to Cooling.
autoHeating	Stimulus. Raised automatically when $\text{currentTemp} < \text{targetTemp} - \text{tempTolerance}$. Transitions from FanOnly to Heating, or from Cooling to Heating.
autoFanOnly	Stimulus. Raised automatically when currentTemp reaches targetTemp . Transitions from Cooling or Heating back to FanOnly.
Timer (1000 ms)	Stimulus. Periodic timer in the Cooling state. Fires every 1000 ms and decrements currentTemp by 1.
Timer (1500 ms)	Stimulus. Periodic timer in the FanOnly state. Fires every 1500 ms and increments currentTemp by 1.
Timer (2000 ms)	Stimulus. Periodic timer in the FanOnly state. Fires every 2000 ms and decrements currentTemp by 1.

Exercise 2: Intruder Alarm

Statechart Diagram:



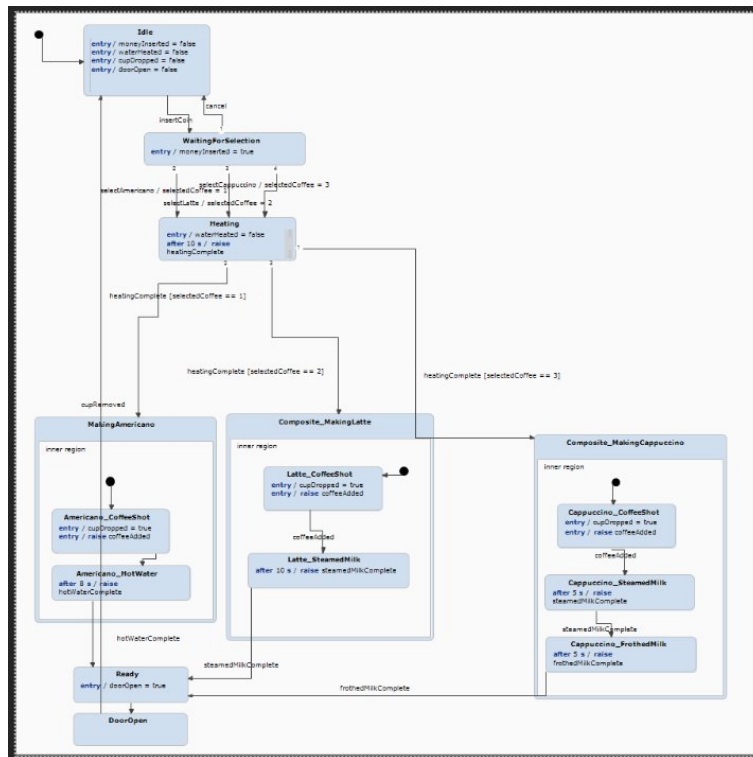
States and Stimuli:

State	Description
Disarmed	The initial state of the system. On entry, pictureStored = false, pictureTransmitted = false, and isRecording = false. Transitions to FullyArmed via 'armFull', to PartiallyArmed via 'armPartial', and can be reached via 'passwordEntered' (raises deletePicture).
FullyArmed	The system is fully armed. On entry, raises alarm signal. Self-transitions on 'intrusionDetected'. Transitions to Beeping via 'timeoutFullArmed', and back to Disarmed via 'passwordEntered' (raises deletePicture).
PartiallyArmed	The system is partially armed. On entry, raises lightKeyboard, raises takePicture, and sets pictureStored = true. After 40 seconds raises 'timeoutPartialArmed'. Self-transitions on 'intrusionDetected'. Transitions to Beeping or back to Disarmed via 'disarm' (raises deletePicture).
Beeping	Alert state indicating a potential intrusion. On entry, raises startBeeping. After 20 seconds raises 'timeoutBeeping'. Transitions to AlarmSounding via 'timeoutBeeping', or back to Disarmed via 'passwordEntered' (raises stopBeeping and deletePicture).
AlarmSounding	The alarm is actively sounding. On entry, raises soundAlarm. Transitions back to Disarmed via 'passwordEntered' (raises stopRecording), or via 'remoteShutoff' (raises stopRecording).
armFull	Stimulus. User activates full arming. Transitions from Disarmed to FullyArmed.
armPartial	Stimulus. User activates partial arming. Transitions from Disarmed to PartiallyArmed.
disarm	Stimulus. User disarms the system. Raised in PartiallyArmed, raises deletePicture and transitions back to Disarmed.

State	Description
passwordEntered	Stimulus. Correct password entered. Raises deletePicture and transitions back to Disarmed from FullyArmed or PartiallyArmed. From Beeping transitions back to FullyArmed. From AlarmSounding raises stopRecording.
intrusionDetected	Stimulus. An intrusion is detected. Causes a self-transition in FullyArmed and PartiallyArmed, keeping the system in its current armed state.
timeoutFullArmed	Stimulus. Raised after the FullyArmed timeout period expires. Transitions from FullyArmed to Beeping.
timeoutPartialArmed	Stimulus. Raised after 40 seconds in PartiallyArmed. Transitions from PartiallyArmed to Beeping.
timeoutBeeping	Stimulus. Raised after 20 seconds in Beeping. Raises stopBeeping and transitions to AlarmSounding.
remoteShutoff	Stimulus. Remote shutdown command. Raises stopRecording and transitions from AlarmSounding back to Disarmed.

Exercise 3: Coffee Machine Dispenser

Statechart Diagram:



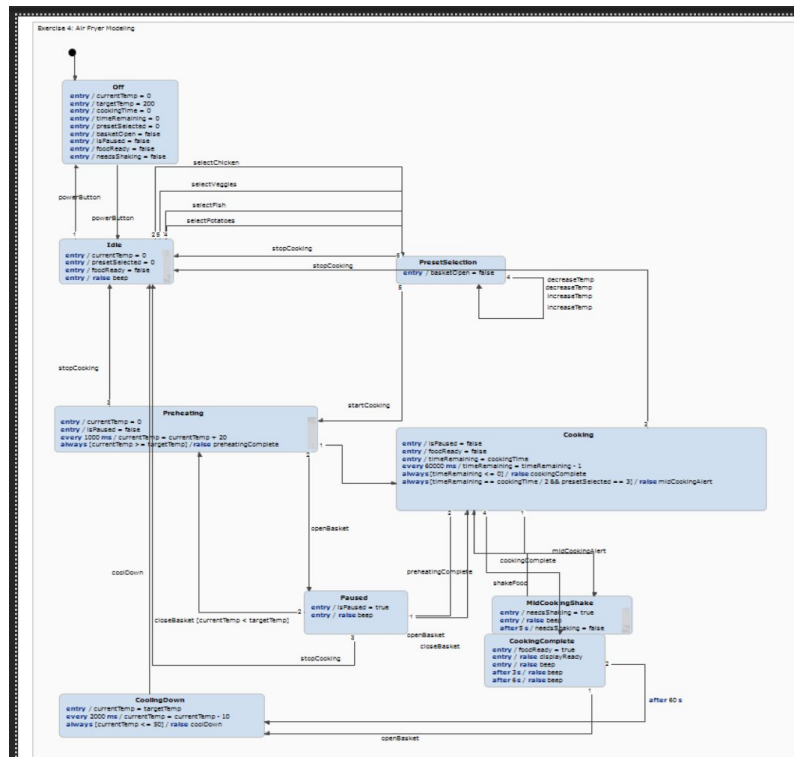
States and Stimuli:

State	Description
Idle	The initial state of the system. On entry, moneyInserted, waterHeated, cupDropped, and doorOpen are all set to false. Transitions to WaitingForSelection via 'insertCoin', and returns here from Ready via 'cupRemoved'.
WaitingForSelection	Waiting for the user to select a coffee type. On entry, moneyInserted = true. Transitions to Heating via 'selectAmericano' (selectedCoffee=1), 'selectLatte' (selectedCoffee=2), or 'selectCappuccino' (selectedCoffee=3). Transitions back to Idle via 'cancel'.
Heating	The system heats the water. On entry, waterHeated = false. After 10 seconds raises 'heatingComplete'. Transitions to MakingAmericano if selectedCoffee==1, Composite_MakingLatte if ==2, or Composite_MakingCappuccino if ==3.
MakingAmericano	Composite state for preparing an Americano. Contains sub-states: Americano_CoffeeShot and Americano_HotWater. Transitions to Ready via 'hotWaterComplete'.
Americano_CoffeeShot	Initial sub-state of MakingAmericano. On entry, cupDropped = true and raises coffeeAdded. Transitions to Americano_HotWater via 'coffeeAdded'.
Americano_HotWater	Second sub-state of MakingAmericano. After 8 seconds raises 'hotWaterComplete'. Transitions out to Ready.
Composite_MakingLatte	Composite state for preparing a Latte. Contains sub-states: Latte_CoffeeShot and Latte_SteamedMilk. Transitions to Ready via 'steamedMilkComplete'.
Latte_CoffeeShot	Initial sub-state of Composite_MakingLatte. On entry, cupDropped = true and raises coffeeAdded. Transitions to Latte_SteamedMilk via 'coffeeAdded'.

State	Description
Latte_SteamedMilk	Second sub-state of Composite_MakingLatte. After 10 seconds raises 'steamedMilkComplete'. Transitions out to Ready.
Composite_MakingCappuccino	Composite state for preparing a Cappuccino. Contains sub-states: Cappuccino_CoffeeShot, Cappuccino_SteamedMilk, and Cappuccino_FrothedMilk. Transitions to Ready via 'frothedMilkComplete'.
Cappuccino_CoffeeShot	Initial sub-state of Composite_MakingCappuccino. On entry, cupDropped = true and raises coffeeAdded. Transitions to Cappuccino_SteamedMilk via 'coffeeAdded'.
Cappuccino_SteamedMilk	Second sub-state of Composite_MakingCappuccino. After 5 seconds raises 'steamedMilkComplete'. Transitions to Cappuccino_FrothedMilk.
Cappuccino_FrothedMilk	Third sub-state of Composite_MakingCappuccino. After 5 seconds raises 'frothedMilkComplete'. Transitions out to Ready.
Ready	Coffee is prepared and ready for collection. On entry, doorOpen = true. Transitions to DoorOpen automatically, returns to Idle via 'cupRemoved'.
DoorOpen	The dispensing door is open awaiting cup removal. Transitions back to Idle via 'cupRemoved'.
insertCoin	Stimulus. User inserts a coin. Transitions from Idle to WaitingForSelection.
cancel	Stimulus. User cancels the selection. Transitions from WaitingForSelection back to Idle.
selectAmericano	Stimulus. User selects Americano. Sets selectedCoffee=1 and transitions to Heating.
selectLatte	Stimulus. User selects Latte. Sets selectedCoffee=2 and transitions to Heating.
selectCappuccino	Stimulus. User selects Cappuccino. Sets selectedCoffee=3 and transitions to Heating.
heatingComplete	Stimulus. Raised after 10 seconds in Heating. Guard determines next state based on selectedCoffee value.
coffeeAdded	Stimulus. Raised on entry to each CoffeeShot sub-state. Triggers progression to the next sub-state.
hotWaterComplete	Stimulus. Raised after 8 seconds in Americano_HotWater. Transitions out of MakingAmericano to Ready.
steamedMilkComplete	Stimulus. Raised after 10 s in Latte_SteamedMilk or 5 s in Cappuccino_SteamedMilk. Transitions to next sub-state or to Ready.
frothedMilkComplete	Stimulus. Raised after 5 seconds in Cappuccino_FrothedMilk. Transitions out to Ready.
cupRemoved	Stimulus. User removes the cup. Transitions from Ready/DoorOpen back to Idle.

Exercise 4: Air Fryer Modeling

Statechart Diagram:



States and Stimuli:

State	Description
Off	The initial state of the system. On entry, currentTemp=0, targetTemp=200, cookingTime=0, timeRemaining=0, presetSelected=0, basketOpen=false, isPaused=false, foodReady=false, needsShaking=false. Transitions to Idle via 'powerButton'.
Idle	System is powered on and waiting. On entry, currentTemp=0, presetSelected=0, foodReady=false, and raises beep. Transitions to PresetSelection via food selection stimuli. Transitions back to Off via 'powerButton'. Returns here from Cooking via 'stopCooking'.
PresetSelection	User is selecting cooking settings. On entry, basketOpen=false. Allows temperature adjustment via 'increaseTemp' and 'decreaseTemp' (self-transitions). Transitions to Preheating via 'startCooking', back to Idle via 'stopCooking'.
Preheating	The air fryer heats up to target temperature. On entry, currentTemp=0 and isPaused=false. Every 1000 ms, currentTemp increases by 20. Raises 'preheatComplete' when currentTemp >= targetTemp. Transitions to Paused via 'openBasket', to Cooking via 'preheatComplete', to Idle via 'stopCooking'.
Cooking	The air fryer is actively cooking. On entry, isPaused=false, foodReady=false, timeRemaining=cookingTime. Every 60000 ms, timeRemaining decrements by 1. Raises 'cookingComplete' when timeRemaining<=0. Raises 'midCookingAlert' when timeRemaining==cookingTime/2 and presetSelected==3.
Paused	Cooking or preheating is paused due to basket opening. On entry, isPaused=true and raises beep. Transitions back to Preheating or Cooking via 'closeBasket', to Idle via 'stopCooking'.

State	Description
MidCookingShake	Mid-cooking shake reminder triggered. On entry, needsShaking=true. After 5 seconds, needsShaking=false. Transitions back to Cooking via 'shakeFood'.
CookingComplete	Cooking has finished. On entry, foodReady=true, raises displayReady and beep. After 3 s raises beep again. After 6 s raises beep again. Transitions to CoolingDown via 'openBasket' or automatically after 60 seconds.
CoolingDown	The air fryer is cooling down. On entry, currentTemp=targetTemp. Every 2000 ms, currentTemp decreases by 10. Raises 'coolDown' when currentTemp<=50. Transitions to Idle via 'coolDown' or 'openBasket'.
selectChicken	Stimulus. User selects the Chicken preset. Transitions from Off or Idle to PresetSelection with chicken cooking profile.
selectVeggies	Stimulus. User selects the Veggies preset. Transitions from Off or Idle to PresetSelection with vegetable cooking profile.
selectFish	Stimulus. User selects the Fish preset. Transitions from Off or Idle to PresetSelection with fish cooking profile.
selectPotatoes	Stimulus. User selects the Potatoes preset. Transitions from Off or Idle to PresetSelection with potatoes cooking profile.
powerButton	Stimulus. User presses the power button. Toggles between Off and Idle.
startCooking	Stimulus. User confirms settings and starts cooking. Transitions from PresetSelection to Preheating.
stopCooking	Stimulus. User manually stops cooking. Transitions from PresetSelection, Preheating, Paused, or Cooking back to Idle.
increaseTemp	Stimulus. User increases the target temperature. Self-transition in PresetSelection.
decreaseTemp	Stimulus. User decreases the target temperature. Self-transition in PresetSelection.
openBasket	Stimulus. User opens the fryer basket. Transitions from Preheating or Cooking to Paused, or from CookingComplete to CoolingDown.
closeBasket	Stimulus. User closes the basket. Transitions from Paused back to Preheating (if currentTemp < targetTemp) or Cooking.
preheatComplete	Stimulus. Raised automatically when currentTemp >= targetTemp. Transitions from Preheating to Cooking.
cookingComplete	Stimulus. Raised automatically when timeRemaining <= 0. Transitions from Cooking to CookingComplete.
midCookingAlert	Stimulus. Raised at halfway point when presetSelected==3. Transitions from Cooking to MidCookingShake.
shakeFood	Stimulus. User shakes the food. Transitions from MidCookingShake back to Cooking.
coolDown	Stimulus. Raised automatically when currentTemp <= 50. Transitions from CoolingDown back to Idle.